Ayush Kumar

Software Engineer | Backend & Full Stack Developer

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Summary

Backend-focused Software Engineer with **2.5+ years** of experience owning and scaling production systems across e-commerce, loyalty, retail, payments, logistics and HR. Core expertise in **Node.js**, **TypeScript**, **REST APIs**, **microservices**, **SQL** and **Redis** on **AWS**, with experience designing high-throughput services supporting **100K+ customers**, **10,000+ concurrent users** and **10 lakh+ daily payout transactions**. Proven track record of designing **caching strategies**, **idempotent payment workflows** and **service decomposition** to improve **API** and **database performance** while modernizing legacy systems into microservices. Strong problem-solving foundation with **671+ LeetCode problems solved** and a **500 Days coding badge**.

Technical Skills

Languages	JavaScript, TypeScript, Java, Python, Go (GoLang)
Backend	Node.js, Express.js, REST APIs, GraphQL, Microservices, Spring Boot
Databases	MySQL, MSSQL, MongoDB, Oracle Database, Redis
Frontend	React.js, Angular, Vue.js, Flutter
Architecture	Distributed Systems, System Design, API Design, Event-Driven Architecture, Caching, Concurrency, Performance Optimization
Cloud & DevOps	AWS, Docker, Kubernetes, Jenkins, GitHub Actions, Nginx, Linux
Security	JWT Authentication, RBAC, OAuth, API Security, OWASP Top 10, VAPT Remediation, Input Validation, Encryption
Integrations	Razorpay, PayU, Worldline, Shopify (REST/GraphQL), SAP, Freshdesk, Shiprocket, Delhivery, Firebase, APNs, Zillion
Tools	Git, Postman, VS Code, IntelliJ IDEA, Android Studio

Professional Experience

Software Engineer Mar 2024 – Present

DS Group *Noida, India*

- Owned end-to-end engineering delivery across **6+ production applications** spanning e-commerce, loyalty, payments, logistics and HR, taking features from technical design through deployment and production support.
- Scaled backend systems to support **100K+ customers** and **10,000+ concurrent users**, with production workloads processing **10L+ daily payout transactions** across payments and loyalty platforms.
- Drove the modernization of legacy **PHP** systems into independently deployable **microservices**, improving release velocity, fault isolation and long-term maintainability across the engineering org.
- Owned secure payment and payout workflows across **Razorpay**, **PayU** and **Worldline**, ensuring reliable transaction processing, reconciliation and fraud prevention at scale.
- Designed and maintained **250+ REST APIs** and third-party integrations spanning commerce, loyalty, payments and logistics workflows.
- Reduced deployment cycle time by **60%** by standardizing CI/CD and release workflows across **AWS**, **Docker**, **Jenkins**, PM2 and Nginx.
- Remediated **150+ high-severity VAPT findings**, raising the security posture of authentication, payment and API layers to **OWASP**-aligned standards.
- Led team-level project execution – breaking down technical work, unblocking teammates and reviewing solutions across multiple concurrent deliverables.
- Partnered directly with product, business and operations stakeholders to translate customer and logistics pain points into scalable engineering solutions, including an **AI-assisted logistics workflow** for OVINO aimed at reducing manual operational effort.

Key Projects

Rajnigandha Loyalty & CRM Platform Mar 2024 – Present

Node.js, React.js, TypeScript, MySQL, Microservices, Redis, Spring Boot, Razorpay, Shopify, SAP

- Designed and decomposed a legacy **PHP monolith** into independently deployable **Node.js microservices** (user management, coupon validation, reward distribution, payments), with clear service boundaries to enable independent scaling and deployment.
- Designed a multi-layer **caching strategy** using **Redis** combined with query indexing and connection pooling on **MySQL**, reducing query execution time by **65%** and improving high-frequency reads by **55%**.
- Engineered **idempotent** reward-redemption and payout workflows with **retry handling**, transaction validation and reconciliation logic to safely process **10L+ daily UPI**/bank-transfer payouts without duplicate disbursement.
- Designed scalable, stateless API layers to support **10,000+ concurrent users**, improving throughput and API performance by **40%+** through backend optimization and efficient data-access patterns.

- Built **Shopify REST/GraphQL** and **SAP** integrations for real-time order, invoice and customer synchronization, and extended CRM workflows with a **Freshdesk** integration that reduced support response time from **48 hours to under 6 hours**.
- Strengthened authentication and authorization using **JWT** and **RBAC**, and remediated **15+ VAPT findings** across API, payment and data-access layers.

Ksheer & OVINO Enterprise Dairy E-Commerce Jul 2024 – Present
Node.js, Angular, TypeScript, MSSQL, Razorpay, PayU, Worldline, Jenkins, Nginx

- Scaled the platform from **2,000+ to 50,000+ customers** while daily GMV increased from approximately **Rs. 20K to Rs. 3L+**, evolving the database and order-processing design to support growth.
- Optimized **MSSQL**-backed **Node.js** APIs through query optimization, indexing and efficient data-access patterns to maintain responsive order, customer and payment workflows under increasing transaction volume.
- Implemented recurring subscription and auto-pay workflows with **retry logic** and failure handling across **Razorpay**, **PayU** and **Worldline**, improving fault tolerance for recurring transactions and refund processing.
- Owned backend and platform engineering for customer-facing dairy e-commerce systems, covering product catalogue, checkout, order lifecycle, subscriptions, delivery scheduling, refunds and administrative workflows.
- Developed **Angular**-based administrative workflows for catalogue, orders, delivery operations and customer management, collaborating with product and operations teams to eliminate manual process bottlenecks.
- Implemented **JWT**-based authentication and **RBAC** across customer, delivery and admin workflows and resolved security issues identified through **VAPT** assessments.
- Built rollback-safe release workflows using Git, **Jenkins**, PM2 and Nginx, troubleshooting application and server issues to minimize production downtime.
- Contributed to an **AI-assisted logistics workflow** for OVINO to improve operational decision-making and reduce manual coordination across delivery processes.

DSphere – Employee Self-Service Platform Aug 2025 – Present
Node.js, React.js, MySQL, REST APIs, JWT, RBAC, PM2, Nginx

- Automated **200+ recurring HR requests per month** through self-service workflows, reducing dependency on manual HR operations and improving request resolution efficiency.
- Designed HR ticketing workflows with **SLA-based status tracking**, approval hierarchies and role-based access for employees, HR, administrators and super administrators.
- Built and maintained secure employee self-service workflows for HR documents, salary slips, Form 16, relieving letters, news feeds and HR ticketing, supporting self-service access to frequently requested information.
- Implemented **JWT authentication** and **RBAC** across sensitive employee-data APIs and handled production deployment, debugging and server configuration using Git, PM2 and Nginx.

Farm Connect – Field Operations Platform Dec 2024 – Present
Flutter, React.js, Node.js, Express.js, MySQL, Firebase, GoLang, SAP, Python

- Built an end-to-end field-operations platform serving **500+ field agents**, pairing **Flutter** mobile apps with a **React** admin portal and **Node.js** backend services for centralized data handling.
- Designed and deployed an **AI-powered disease detection** feature that classifies crop issues from photos taken in the field and suggests treatment actions, helping agents make faster, more accurate on-site decisions.
- Implemented **SAP** integration to streamline **inventory management**, **purchase order (PO) processing**, and **document management**, cutting manual effort and keeping field and back-office records consistent.
- Secured the platform with **JWT-based authentication** and **RBAC**, and built **event-driven** push notifications using **Firebase Cloud Messaging** along with **GPS-based tracking** to monitor field activity.
- Tuned backend queries and API workflows for better performance, and remediated **VAPT** findings across authentication and API layers ahead of production release.

Education

GLA University 2020 – 2024
B.Tech. in Computer Science and Engineering | CGPA: 7.88/10 *Mathura, India*

Achievements

- **Star Performer – 2025-2026:** Recognized for engineering contribution, ownership and delivery impact at DS Group.
- **LeetCode:** Solved **671+ problems** across Data Structures, Algorithms, SQL and problem-solving; earned the **500 Days Badge** and multiple annual/daily challenge badges.

Certifications

- Machine Learning, Artificial Intelligence & Software Development Lifecycle – Udemy
- IoT / IIoT Communication – GLA University
- Data Science & Analytics – GLA University